RANCH OS

The Operating System for Ranch Operations

One System. Every Ranch. Every Day.

Private, modular operations for land, livestock, equipment, finances, health, and knowledge.

MISSION

Empower ranchers with a complete operating system that manages every aspect of ranch operations with offline-first reliability and AI-enhanced intelligence.

CORE PRINCIPLES

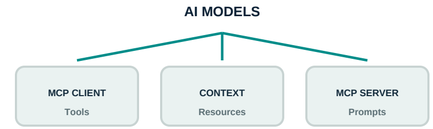
- ✓ Offline-first always
- ✓ AI-augmented, not replaced
- ✓ Open system and community driven
- ✓ Privacy and data ownership
- ✓ Modular and extensible
- ✓ Built for ranchers, by ranchers

WHAT IS RANCH OS?

1. Ranch OS is the operating system
2. Ranch operations modules
3. AI intelligence layer
4. Hardware integration
5. Home automation
6. MCP model client

All working together seamlessly.

MCP ARCHITECTURE



RANCH OS PLATFORM

OpenClaw is the supporting technology foundation

CLIENT APPLICATIONS

Cross-platform access

- **Mac** — Management
- **iPad** — In the office
- **iPhone** — In the field
- **Web portal** — Private access
- **Apple TV** — At a glance

Secure access wherever work happens.

APPLICATION MODULES

Property Manager	Livestock Manager	Field Manager	Inventory Manager	Finance Manager	Home Automation	RanchBrain	More Modules
Assets and maintenance	Identity, health, breeding	Land, pasture, location	Supplies and inventory	Records and costs	Sensors and routines	Knowledge and insight	Modular extensions

OPENCLAW CORE PLATFORM

Identity	Tenant context	Authorization	RLS	Data sync	Local database	Tasks and alerts	Settings
Authentication	Fail-closed	Scoped roles	Data isolation	Offline-first	PostgreSQL	History	Preferences

AI INTELLIGENCE LAYER

AI router	Apple FM	oMLX / Qwen	MCP client	RAG system	Vector DB
Model selection	On-device	Local models	Context	Knowledge	Qdrant

DATA LAYER

Operational data	Reference data	Documents	Knowledge base	Embeddings	Audit history
Animals, land, work	Assets, equipment	Manuals, PDFs	Curated insights	Semantic search	Change tracking

HARDWARE INTEGRATION LAYER

Field mounts	Bluetooth scales	RFID readers	Barcode scanners	IoT sensors	GPS trackers	More devices
Mobile use	Weights	Animal tags	Inventory	Health and environment	Location	Extensible

AI MODELS

LOCAL REASONING

oMLX / Qwen: knowledge, reasoning, records, retrieval.

APPLE ON-DEVICE

Foundation Models: summaries, alerts, classification.

LOCAL FALLBACK

Hermes 8B, Qwen 9B/14B, approved local models.

Tenant-authorized, provenance-aware, policy-bounded.

SYNC & INFRASTRUCTURE (OFFLINE-FIRST)

Client devices ↔ **Local database** ↔ **Gateway services** ↔ **Field services**

ECOSYSTEM INTEGRATIONS (LOCAL-FIRST)

QuickBooks local sync	Label printing	Weather API
Camera systems	Maps and GIS	More integrations

TECH STACK (LOCAL, MODULAR)

Client: SwiftUI / React
Backend: Python FastAPI
Data: PostgreSQL + Qdrant
AI: oMLX, Apple FM, local models
Protocol: MCP

BUILT FOR RANCHERS

Understands ranching, works in the real world, and stays private.

- ✓ Complete ranch management
- ✓ Tenant-safe for multiple ranches

KEY BENEFITS

- ✓ Works offline in remote locations
- ✓ Private data stays private

THE VISION

One private operating system for ranchers - built for today, ready for a cloud-enabled future.

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One system. Every ranch. Every day.